

# Aleena Varghese

Software Engineer — Full-Stack, APIs & AI/ML

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## PROFILE

Software engineer building **production systems end to end** — full-stack platforms, REST APIs, AI/ML and Google Cloud infrastructure. Founding engineer of **CelerSCET**, a campus-wide ERP (90+ modules, ~500k lines of code) that runs admissions, enrollment, academics and administration for an engineering college, with **1,600+ commits and 680+ pull requests merged** as platform lead. MSc in Artificial Intelligence & Machine Learning; IEEE-published GAN research. Focused on system design, data integrity, automated testing and reliable cloud deployments.

## PROFESSIONAL EXPERIENCE

### Software Engineer · Full-Stack Developer

11/2025 – Present

*CelerSCET — Sahrdaya College of Engineering and Technology, Kerala*

- **Architected and bootstrapped a campus-wide ERP** serving the full student lifecycle in production — React 18/Vite SPA (PWA), Node.js/Express REST API (~79 route segments), Supabase PostgreSQL with row-level security, deployed on Google Cloud Run.
- **Engineered the admission platform end to end** — multi-step applications, OTP applicant authentication, quota- and degree-aware validation, Razorpay payment integration, QR verification, branded PDF exports and analytics — processing **1,500+ applications and 500+ admissions** in its first cycle.
- **Designed the student provisioning pipeline** — batch creation, class allocation, SR-number generation and institutional email — as a dependency-ordered workflow that converts admitted applicants into live student accounts.
- **Implemented platform-wide authorization and session security** — a 3-layer permission engine with relationally scoped rights (HoD, class teacher, coordinators), device limits and enforceable session revocation — plus registry-driven notifications, a generic approvals engine and an audit trail with restorable deletes.
- **Own CI/CD and cloud infrastructure** — Cloud Build → Cloud Run (staging + production), Artifact Registry, Cloud Storage, Secret Manager, Cloud Logging; reduced build times from 45+ minutes to ~4 and eliminated stale-deploy caching failures.
- **Lead integration for a 5-engineer team** — 680+ pull requests reviewed and merged; top contributor to a 275-file automated test suite (~72k lines) guarding system invariants.

### AIML Executive · Full-Stack Developer (APIs & AI/ML)

01/2025 – 12/2025

*SenseSys Pvt. Ltd. — Bangalore, Karnataka*

- **Led an 8-member agile team** as project lead for **Hosting Cloudly**, a cloud hosting platform — developed the React dashboard for uploads, deployment tracking and performance metrics, and integrated REST APIs for authentication, deployment status and log visualization.
- **Directed the ZeroCodeML frontend revamp** — a no-code machine learning platform with automated EDA and model suggestions — building modular React components for model setup, data upload and training progress.
- **Optimized UI load performance by ~30%** through lazy loading, component optimization and asset compression; integrated Supabase-backed authentication and data services.
- Ran sprint planning, code reviews and Git workflows across the team.

**ML Engineer Intern, Prodigy Infotech** (04/2024 – 05/2024) · **Python Developer Intern, Psyliq** (05/2024 – 06/2024) · **Graphic Designer, NextU Empower** (09/2024 – 12/2024)

## PROJECTS

### CelerSCET — Digital Campus ERP (production)

2026

*React · Node.js · Express · Supabase/PostgreSQL · Google Cloud*

- 90+ module ERP ecosystem covering admissions → provisioning → academics → administration; live at [sahrdaya.celerscet.com](https://sahrdaya.celerscet.com) (detailed under Professional Experience).

### Hosting Cloudly — Cloud Hosting Platform

2025

*React · FastAPI · Supabase · Docker*

- End-to-end hosting platform (domains, Linux hosting, WordPress deployment) with dashboards for deployment tracking and monitoring; led the 8-member build team at SenseSys.

## ZeroCodeML — No-Code Machine Learning Platform

2025

React · Flask · scikit-learn

- Upload a CSV and train models without writing code — automated EDA, model suggestions and training progress; led a 3-member team through the V1.0 launch.

## HospSurv — Hospital Feedback Platform

2025

React · JavaScript · Supabase

- Patient survey system with reward coupons, gamified survey flows and real-time admin dashboards for response tracking.

## SKILLS

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**Languages:** JavaScript (ES6+), Python, SQL, HTML5, CSS3

**Frontend:** React 18, Vite, React Router, MUI, Tailwind CSS, Progressive Web Apps

**Backend & APIs:** Node.js, Express, REST API design, JWT authentication, Flask, FastAPI, Nodemailer, Razorpay integration

**AI / ML:** PyTorch, TensorFlow, scikit-learn, GANs, Pandas, NumPy, Natural Language Processing

**Databases:** PostgreSQL, Supabase (RLS, Auth, Realtime), Redis, MySQL, SQLite

**Cloud & DevOps:** Google Cloud Platform (Cloud Run, Cloud Build, Cloud Storage, Artifact Registry, Secret Manager, Cloud Logging), Docker, CI/CD, Git/GitHub, Postman

**Practices:** Automated testing (node:test), code review, technical documentation, agile delivery

## PUBLICATIONS

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### “Enhancing Satellite Imagery with GAN-based Cloud Removal” — IEEE

07/2025

Published in *IEEE Xplore* | DOI: [10.1109/ICECCC60120.2025.11063844](https://doi.org/10.1109/ICECCC60120.2025.11063844) — presented at ICECCC 2025, CMR Institute of Technology, Bengaluru

- Engineered a GAN architecture (U-Net generator, ResNet discriminator, Perlin-noise cloud synthesis) that improved spatial accuracy and clarity by ~40% over baseline models.
- Utilized tensor-level optimizations and GPU-based training to accelerate convergence; benchmarked on EuroSAT with PSNR validation for Earth-observation preprocessing.

## EDUCATION

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### MSc in Artificial Intelligence and Machine Learning

2023 – 2025 | Bangalore, Karnataka

CHRIST (Deemed to be University)

### Bachelor of Computer Applications

2020 – 2023 | Rajkot, Gujarat

Christ College — Outstanding Student Award, Dept. of Computer Science

## CERTIFICATES

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Deloitte Data Analytics Job Simulation · AWS Cloud Foundations · Machine Learning with Python · Natural Language Processing · Prompt Engineering for Generative AI · Introduction to Power BI · Data Engineering · Introduction to LLMs · Ethical AI

## LEADERSHIP & CONTRIBUTOR EXPERIENCE

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- **Project Lead — Hosting Cloudly & ZeroCodeML:** led an 8-member team at SenseSys building an AI-powered hosting platform, and a 3-member team at CHRIST University building a no-code ML tool.
- **Media Head — Gateways 2023:** directed a 30-member team covering branding, content strategy and social media for a national tech fest; produced 50+ event visuals, increasing digital reach by 80%.
- **Workshop Facilitator — Canva & Storytelling:** conducted design and visual-storytelling workshops for 100+ students, including a 2-day national session at Illuminaire.
- **Volunteer — ICSAC 2024:** committee member, International Conference on Sustainable Advanced Computing · **Student Coordinator — TechArena:** digital strategy for Christ College's national IT fest.

## LANGUAGES

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English | Malayalam | Tamil | Hindi | Gujarati